Class-Aligned and Class-Balancing Generative Domain Adaptation for Hyperspectral Image Classification

Jie Feng, *Senior Member, IEEE*, Ziyu Zhou, Ronghua Shang, *Senior Member, IEEE*, Jinjian Wu, *Senior Member, IEEE*, Tianshu Zhang, Xiangrong Zhang, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— The task of hyperspectral image (HSI) classification is fundamental and crucial in HSI processing. Currently, domain adaptive methods have become a research hotspot in HSI classification. However, most domain adaptive methods ignore the class alignment in different domains. Additionally, HSIs have the characteristics of category imbalance and complex spatial–spectral distribution, which restricts the adaptation performance in HSIs. To address these problems, a class-aligned and class-balancing generative domain adaptation (CCGDA) method is proposed for HSI classification. The architecture of CCGDA is designed by using the classifier, domain discriminator, sampler, and two weight-sharing generators. In the classifier, split-level capsule network (CapsNet) is constructed by extracting rich spatial information of shallow layer and spectral features of deep layer with equivariant characteristic. Then, the classifier provides the pseudo-label of samples in the target domain. To prevent the generators from mode collapse caused by category imbalance, the sampler is designed. It samples and resamples the samples of the target domain in an adaptive proportion according to the statistical calculation through confidence and distribution of pseudo-labels. Finally, a novel class-aligned domain adversarial loss is defined to jointly optimize the generators and discriminator. It incorporates the class shift adjusting and adaptive sampling for the samples of the target domain to better adapt the discriminant boundary of the classifier to the target domain. Experiments on benchmark HSI datasets verify the superiority of the proposed method for domain adaptive classification.**

***Index Terms*— Capsule network (CapsNet), class imbalance, domain adaptation (DA), generative adversarial network (GAN), hyperspectral image (HSI).**

Manuscript received 14 September 2023; revised 28 December 2023; accepted 5 February 2024. Date of publication 20 February 2024; date of current version 28 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271374, Grant 62176200, Grant 62077038, and Grant 62176196; in part by the State Key Program of National Natural Science of China under Grant 61836009; in part by the Natural Science Basic Research Program of Shaanxi under Grant 2022GY-065 and Grant 2022JC-45; and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX23047. *(Corresponding author: Jie Feng.)*

The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: jiefeng0109@163.com; zzyzhouziyu@outlook.com; rhshang@mail.xidian.edu.cn; jinjian.wu@mail.xidian.edu.cn; zts.xidian@gmail.com; xrzhang@mail.xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3367765

NOMENCLATURE

x_s	Source data.
x_t	Target data.
x_f	Fake date.
$\tilde{x}_t$	Resampled target data.
y_s	Label of source data.
y_t	Label of target data.
$\hat{y}_s$	Prediction vector of x_s.
$\hat{y}_t$	Prediction vector of x_t.
$\hat{y}_f$	Prediction vector of x_f.
$\tilde{y}_t$	Prediction vector of $\tilde{x}_t$.
z	Random noise.
Θ_G	Parameters of classifier.
Θ_C	Parameters of discriminator.
Θ_D	Parameters of generator.
SD	Source data.
TD	Target data.
FD	Generated fake data.

I. INTRODUCTION

HYPERSPECTRAL image (HSI) is a kind of special remote sensing source with high spectral resolution. It has dozens or even hundreds of narrow spectral bands and abundant spatial information as well. Due to their high dimensionality, HSIs can represent a broader range of objects and have irreplaceable advantages in various fields such as land cover classification [1], [2], object detection [3], [4], anomaly detection [5], and so on [6], [7].

Image classification is an important application scenario for HSIs. Over the past few decades, numerous traditional machine learning-based algorithms have been developed, such as support vector machine (SVM) [8] and k-nearest neighbor (KNN) [9]. In recent years, HSI classification algorithms based on deep learning have emerged and achieved remarkable performance, becoming the mainstream in this field irreversibly [10], [11], [12], [13]. Most deep learning-based image classification algorithms rely on labeled data for supervised training heavily. Moreover, due to the presence of domain shift, the classification model trained on a specific dataset (source domain) can hardly be transferred to another dataset (target domain), which severely limits the usage of

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

the deep learning model. If the target dataset is fully labeled or partially labeled, we can easily train a new model in a supervised way or fine tune the existing model. However, the labeling process of HSIs consumes excessive labor and time. In most cases, we are faced with completely unlabeled TD, and domain adaptation (DA) is suitable for this situation to solve the problem of model transfer. This task can also be described as unsupervised DA.

The most intuitive solution to DA is to eliminate the domain shift, that is, to achieve the purpose of model transferring by narrowing the inconsistency between the two domains [14]. Most methods focus on addressing the data distribution shift, aligning the target and source domains by reducing domain discrepancy [15], [16], [17] or generating the intermediate transition domain [18], [19], to extract domain-invariant features for classification. Among these methods, the generative DA method inspired by generative adversarial network (GAN) [20] can perform instance-level domain translation, showing stronger learning ability for domain-invariant features, and expanding the training data of the classification model. However, most generative DA methods only consider the overall distribution in target domain when generating synthetic samples, without distinguishing the distribution difference between classes. As a result, the classification boundaries trained on the source domain cannot be well applied to the target domain, because the corresponding classes are misaligned. Some self-training-based method like class-balanced self-training (CBST) [21] would balance the class distribution by iteratively filtering the majority class according to the pseudo-label, so as to solve the problem of class distribution shift and make the probability distribution of the target domain more balanced. However, the filtering operation may reduce the sample richness and limit the generalization ability of the classifier.

In recent years, some scholars have introduced domain adaptive techniques into the field of HSI classification. Zhang et al. [22] proposed a topological structure and semantic information transfer network (TSTnet), to extract domain-invariant topological features and semantic features by combining convolutional neural network (CNN) and graph convolutional network (GCN). It uses maximum mean discrepancy (MMD) to measure the difference between the source and target features extracted by CNN, and uses graph optimal transmission (GOT) in GCN to measure the topological difference between the source and target features. Jia et al. [23] proposed gradient feature-oriented 3-D DA (GF-3DDA) from the perspective of 3-D Gabor filter. They designed 3-D correlation alignment technique for DA. Later, Huang at al. [24] proposed a two-branch attention adversarial DA (TAADA) network based on the idea of maximum classifier discrepancy (MCD) [25]. TAADA reconstructs the feature extractor in MCD into a dual-branch structure to better extract the spatial–spectral features of HSIs. Besides, the instance-to-instance contrastive learning framework based on category matching (CLCM) [26] proposed by Ning et al. and the refined prototypical contrastive learning network for FSL (RPCL-FSL) [27] proposed by Liu et al. use the method of comparative learning to constrain the feature extraction network and extract domain-invariant features.

Although these methods have made specific improvements for HSIs, most of them overlook the issue of category alignment. The unknown class distribution shift in the target domain makes the adaptation uncontrollable. Especially, the imbalanced class distribution in the target domain leads to the class preference of adaptation. In HSIs, the sample ratio of majority classes and minority classes can even reach thousands of times, which deteriorates the classification performance on minority classes. Furthermore, there is the problem of "same object with different spectrum" in HSIs. To alleviate this problem, the existing methods consider the joint spatial–spectral features for classification, but it is difficult to take into account the generalization ability for such refined features simultaneously. How to extract highly generalized spatial–spectral features is worth studying.

To address the above issues, a class-aligned and class-balancing generative DA (CCGDA) method is proposed for HSI classification. CCGDA consists of the classifier, domain discriminator, sampler, and two weight-sharing generators. In classifier, a novel split-level capsule network (CapsNet) is constructed to extract spatial and spectral features in HSIs under different receptive fields using a staggered structure. The equivariant characteristic brought by split-level CapsNet makes the classifier have highly generalized ability for the samples of the same category with different spectra. The classifier predicts the pseudo-labels of TD and FD synthesized by the generator. Then, the data sampler will sample and resample the TD according to the class distribution and confidence of pseudo-labels, so that rebalance the class distribution of TD and eliminate the interference caused by undertrained classifier. Finally, the resampled TD is sent into the domain discriminator together with FD for adversarial training. To achieve joint adversarial optimization, a novel class-aligned domain adversarial loss is defined. It quantifies the category correlation between the samples from TD and FD, and dynamically adjusts the optimization direction of class and distribution alignment of the generator.

The main contributions of this article are summarized as follows.

1) The novel DA framework overcomes the problem of unknown class distribution shift in target domain, and alleviates the extreme adaptation preference caused by class imbalance. It can also extract highly generalized spatial–spectral features, making the DA technique well suitable for HSI classification tasks.
2) In CCGDA, a novel class-aligned domain adversarial loss can guide the generator to learn the target domain distribution with class distinction by quantifying the category correlation between unaligned TD and FD. The data sampler prevents the mode collapse that might occur in the generator due to the unknown class imbalance.
3) It is first discovered that the equivariance of the CapsNet can enhance the generalizability of features in the DA for HSI classification. A split-level CapsNet can extract joint spatial–spectral equivariant features, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

contributes to achieving a good balance between the fine-grained spatial–spectral features for classification and the generalizable features for DA.

The remainder of this article is constituted by the following four sections. First, a brief introduction to DA, GAN, and CapsNet is presented in Section II. Then, the details of the proposed CCGDA are described in Section III. In Section IV, we present the experimental results on three benchmark datasets. Finally, Section V ends this article with our conclusion and thoughts on this work.

II. Background

A. Domain Adaptation

After more than ten years of research and development, the existing domain adaptive methods can be divided into four branches: statistic based, adversarial based, generative based, and self-training based.

1) Statistic-Based Methods: The statistic-based methods often train a network that can extract domain-invariant features by measuring and reducing the discrepancy between the features of two domains, in order to align the data distributions of the source domain and the target domain in a specific feature space. The essential difference between these methods lies in the design of their statistical metric functions. Commonly used metric functions include MMD [28], multikernel MMD (MK-MMD) [29], and covariance [30]. As early as 2010, Pan et al. proposed transfer component analysis (TCA) [28], using MMD to find transferable components in the reproducing kernel Hilbert space. This method was further applied in deep convolutional neural network (CNN) as a kind of domain confusion loss in deep domain confusion (DDC) [16], which was proposed by Tzeng et al. in 2014. Later, to compensate for the instability of single kernel functions and guarantee the unbiased deep features, Long et al. proposed deep adaptation network (DAN) [17] by using multikernel MK-MMD in CNN. Sun et al. proposed deep correlation alignment (deep CORAL) [31], in which the covariance matrices were used to measure the distance between the source and target features. Based on the idea of statistic metric, some researchers have also conducted extended research on HSI classification tasks, such as the TSTnet [22], GF-3DDA [23], and discriminative cooperative alignment (DCA) [32].

2) Adversarial-Based Methods: The adversarial-based methods usually use domain discriminators instead of statistical metrics to evaluate the distance between two domains. When the discriminator cannot distinguish whether the feature comes from the source domain or the target domain, the distribution of the two domains is aligned in the feature space. One of the most representative methods is domain adversarial training of neural network (DANN) [33]. The adversarial discriminative DA (ADDA) [34] proposed by Tzeng et al. also follows this idea. Long et al. proposed conditional domain adversarial networks (CDAN) [35] conditioning the domain discriminator with semantic and predictive features, and the impact of the black box process of the discriminator on domain confusion performance was first noticed. Recently, a spatial–spectral weighted adversarial DA (SSWADA) network [36] proposed by Huang et al. employs an idea of weighted adversarial discrimination to align the feature distribution of source and target domain, which also belongs to this type of method. There are also some other adversarial-based methods that take the idea of MCD [25]. These methods capture domain-invariant features through minimizing the discrepancy between two classifiers (one for the source domain and one for the target domain). The adversarial learning between the classifier and the feature extractor, such as confident learning-based DA (CLDA) [37], TAADA [24], and unsupervised joint adversarial DA (UJADA) [38], achieves preeminent adaptation performance.

3) Generative-Based Methods: Compared with the adversarial-based methods, the generative-based methods draw more inspiration from GAN [20]. Many GAN-based algorithms, including CoGAN [19], DualGAN [39], and CycleGAN [40], learn invariant features between the two domains through domain translation and can be used for domain adaptive classification. Bousmalis et al. [18] proposed pixel-level DA (PixelDA), by generating an intermediate domain with both category features of SD and distribution characteristics of TD, the classifier in PixelDA is trained on a more diverse dataset to achieve high robustness on target domain. Some scholars also use encoder–decoder networks to extract source features and decode it into target domain samples, forcing the network to learn domain-invariant features [41].

4) Self-Training-Based Methods: The self-training-based methods that emerged recently incorporated the idea of semi-supervised learning into DA tasks. This branch of methods improves model prediction certainty by gradually adding high-confidence unlabeled TD samples to the training set. These methods allow the model to improve robustness on target domain while ensuring correct classification on source domain. The above-mentioned CLDA [37] actually combines the self-training technique, which filters out trusted samples based on the output of the pseudo-labels by the classifier to supervise the target domain classification. In addition, there are some methods that take into account the class imbalance in self-training sample selection process to reduce the major categories preference of the pseudo-label generation [21], [42].

B. Generative Adversarial Network

GAN [20] was proposed by Goodfellow et al. in 2014, which consists of a generator and a discriminator. The essence of adversarial training in GAN is to play minmax game between the generator and discriminator. The generator would generate a fake sample to cheat the discriminator, while the discriminator would try to distinguish real and fake. This process can be expressed by the following formula:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log D(x)\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log(1-D(G(z)))\right] \quad (1)$$

where G represents the generator, D represents the discriminator, x is the real sample that follows the p_{data} distribution, and z is a random noise that follows the p_z distribution. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

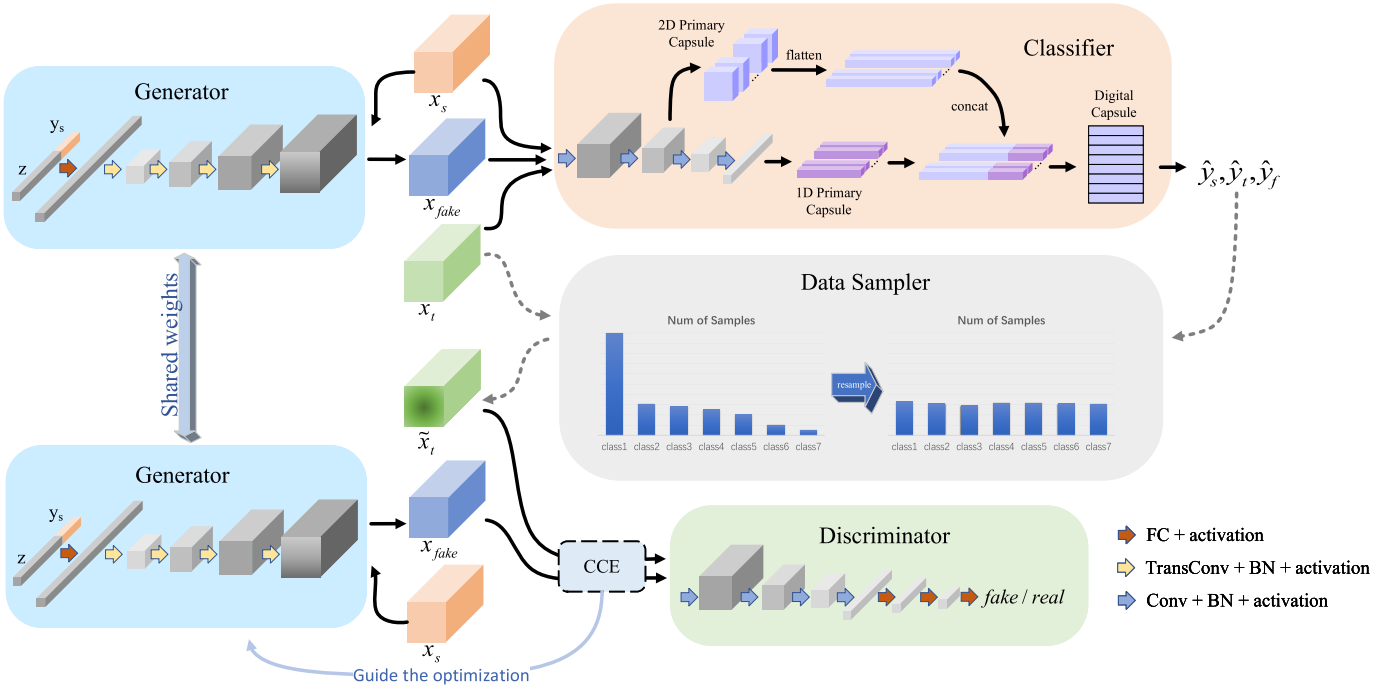


Fig. 1. Flowchart of the proposed method. The framework consists of two generators, a classifier, a discriminator, and a data sampler. CCE refers to the class correlation evaluation, which works in the training process of the second generator.

discriminator aims to correctly classify the real sample x and the fake sample $G(z)$. It assigns a value of 1 to the real sample and a value of 0 to the fake sample to maximize the loss function $V(D, G)$. While the generator aims to confuse the discriminator to classify the fake samples $G(z)$ into real ones, so as to minimize the loss function. In order to achieve this goal, the generator has to learn the real data distribution p_{data}, and let the distribution of fake samples approaches the p_{data} gradually. The alternating training is used during the training process to sequentially improve the performance of the generator and discriminator during the minmax gaming.

C. Capsule Network

CapsNet [43] was proposed by Sabour et al. Its original intention is to solve the defect of low robustness of CNN to image rotation. In traditional CNN, the output of each convolution is a scalar, so it can only answer the question of "yes or no," but cannot describe other attributes, which is an important reason for its poor robustness. While the output of each capsule in CapsNet is a vector, and each parameter in the vector represents a specific attribute such as position, size, and orientation. This makes the output of the capsule equivariant. That is, if a certain attribute changes, the vector will change accordingly.

The CapsNet is constructed by the PrimaryCaps layer and the DigitCaps layer. The PrimaryCaps contains a series of parallel convolutions, where each convolution is called a capsule. The output of these capsules is composed of vectors as the output of PrimaryCaps. The DigitCaps carries out the weight mapping on all the vectors outputted by PrimaryCaps, and maps these vectors into a number of vectors with a specified length of digits, and each vector represents the classification information of a category. The way of updating the mapping parameters from PrimaryCaps to DigitCaps is different from that of backpropagation, and it is completed through a process called the dynamic routing.

At the same time, CapsNet uses the length of the vector to represent the classification probability, and the length is limited to the range of [0, 1]. Because the value of each bit in the vector is limited to [0, 1] by the squash function below

$$v = \frac{||s||^2}{1 + ||s||^2} \cdot \frac{s}{||s||} \tag{2}$$

where s is the original output vector and v is the normalized vector. So, CapsNet can also obtain the invariance of CNN, that is, the probability of being classified into a certain category is always constant for the same target.

For the data with rich spectral sequence information like HSIs, this kind of network with both equivariance and invariance has the potential to solve the problem of the "homogeneous." The "homogeneous" problem for HSI is usually manifested as spectral drift caused by environmental temperature and other factors, that is, the reflection value of an object at a certain wavelength may be different under different situations. But the relative relationship between bands is mostly unchanged when viewed along the horizontal wavelength axis, which means the value in a certain band is always higher or lower than some specific bands. So, it is possible to alleviate the "homogeneous" problem by improving the spectral robustness of the classifier. In the CapsNet, the PrimaryCaps ensures the equivalence of features and can describe the spectral drift, while the DigitCaps ensures the invariance of classification, so the features with spectral drift

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

still can be correctly classified. In other words, it has better spectral robustness.

III. Methodology

In this section, we will introduce the network structure, calculation process, and principle of the proposed method in detail. The flowchart of the proposed CCGDA is shown in Fig. 1. The framework mainly consists of four parts: two generators, a classifier, a discriminator, and a data sampler. The first generator synthesizes the FD through the label of SD and a random noise, and these FD will be sent into the classifier together with real SD for supervised training. The trained classifier predicts the pseudo-label of both FD and the TD. Then the data sampler samples and resamples the TD according to the class distribution and confidence of pseudo-label, so that rebalances the class distribution of TD and eliminates the interference caused by undertrained classifier. Finally, the resampled TD will be sent into the domain discriminator together with FD. There are two groups of joint optimization in the training process, namely the joint optimization between generator and classifier, and the joint adversarial optimization among the generator, the sampler, and the discriminator. Note that the class distribution in TD is unknown and unaligned with FD. It is improper to optimize the generator according to the discrimination result directly. So that a novel class-aligned domain adversarial loss is defined to quantify the category correlation between unaligned TD and FD. Benefiting from the data sampler, the optimization will not encounter mode collapse caused by the majority classes' preference.

A. Classifier With Split-Level CapsNet

As a classifier, the proposed split-level CapsNet has a staggered structure to handle the spatial and spectral information under different receptive fields. Through a 2-D-PrimaryCaps layer, it can extract the spatial features from the shallow layer, where the spatial semantics are initially extracted and the spatial physical information is still reserved. And a 1-D-PrimaryCaps layer is used to extract the spectral information in deep layer, where the spatial information is highly compressed and spectral information is revealed. Then, the DigitCapsules layer is shared by two PrimaryCaps layers to fuse the spatial spectral features and output the prediction vectors of each class, as shown in Fig. 2.

The main body of the classifier is a CNN with four ConvBlocks. Each block contains a convolution layer, a batch normalization (BN) layer, and a activation layer. The features output by the second ConvBlock and the last ConvBlock will be sent into the split-level CapsNet. This process can be expressed as the following formula:

$$\hat{y} = \mathrm{dc}(f), \quad f = \mathrm{squash}((\mathrm{PC}_{1d}(f_{1d}); \mathrm{flat}(\mathrm{PC}_{2d}(f_{2d})))) \tag{3}$$

where f_{1d} and f_{2d} represent the features output by the last ConvBlock and the second ConvBlock, PC_{1d} and PC_{2d} represent the 1-D PrimaryCapsules layer and 2-D PrimaryCapsules layer, and dc represents the DigitCapsules layer. Since $\mathrm{PC}_{2d}(f_{2d})$ is a series of 2-D patches, it must be reshaped into a series of vectors before it can concatenate with $\mathrm{PC}_{1d}(f_{1d})$. The $(;)$ means the operation of concatenating, and $\mathrm{flat}(\cdot)$ means the operation of flattening. The $\mathrm{squash}(\cdot)$ is the squash function mentioned in (2).

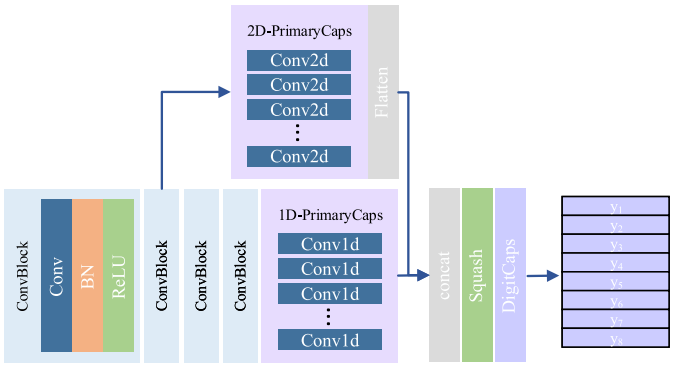


Fig. 2. Structure of classifier with split-level CapsNet.

Different from the way that spatial information and spectral information are split into two independent branches in most HSI classification networks, which need different inputs for each branch (vector for the spectral branch and patch for the spatial branch). The split-level CapsNet can process the spatial and spectral information through a single input in a whole structure.

1) Dynamic Routing: The role of PrimaryCapsules layer is similar to the convolution layer, and the difference is to realize the "vectors in vectors out." A capsule is composed of many convolutions. The role of DigitCapsules layer is similar to the fully connection layer. All the features from PrimaryCapsules are fully connected and mapped into C prediction vectors through a weight matrix and dynamic routing, where C is the number of classes. The process of dynamic routing is formulated as follows:

$$\hat{u}_{j|i} = W_{ij} f_i \tag{4}$$

$$s_j = \sum_i^n c_{ij} \hat{u}_{j|i} \tag{5}$$

where the input vector f_i is multiplied by a weight matrix W_{ij}. The W_{ij} here can be learned in the training process, while the coupling coefficients c_{ij} are determined by the dynamic routing

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_k^C \exp(b_{ik})} \tag{6}$$

$$b_{ij} \leftarrow b_{ij} + \hat{u}_{j|i} v_j. \tag{7}$$

All the values of b_{ij} are initialized to 0. Thus, all the values of c_{ij} are same at first, then these values will be updated by (7). In (7), v_j is squashed from s_j. As the cosine distance between v_j and $\hat{u}_{j|i}$ is smaller, their correlation is higher. In this case, the b_{ij} increases, which leads to the growth of the coupling coefficients c_{ij}.

2) Margin Loss: We use the margin loss instead of cross-entropy (CE) loss to optimize the classifier. It shows better adaptability than CE loss, which we will prove in Section IV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

The margin loss function is calculated as follows:

$$L_{\text{margin}}(\hat{y}, y_s) = \sum_{i}^{C} \Big(T_i \max\big(0, \alpha^+ - ||\hat{y}_i||_2\big)^2 + \lambda(1 - T_i) \max\big(0, ||\hat{y}_i||_2 - \alpha^-\big)^2 \Big) \tag{8}$$

$$L_C = L_{\text{margin}}(\hat{y}_s, y_s) + L_{\text{margin}}(\hat{y}_f, y_s) \tag{9}$$

where $\hat{y}_i$ is the prediction vector of the ith DigitCapsule, y_s is the one-hot label of SD, T_i is the digit value in y_s, $T_i = 1$ if the sample belongs to the ith class, and $||\cdot||_2$ refers to the norm L_2. α^+ and α^- are the upper and lower bounds of margin, respectively. They restrict the digit value of the belonging category in the vector to be as high as possible, while the values of other digits are as low as possible. λ is a weight factor. Usually, $\alpha^+ = 0.9$, $\alpha^- = 0.1$, and $\lambda = 0.5$ are selected, following the suggestion in [43]. $\hat{y}_s$ and $\hat{y}_f$ are the prediction vectors of TD and FD, which share the same label y_s and force the generator to learn the category characteristics in source domain.

B. Entropy-Constrained Class-Balancing Data Sampler

After the classifier is trained, we can obtain $\hat{y}_s$, $\hat{y}_t$, and $\hat{y}_f$, which are the prediction vectors of SD, TD, and FD, respectively. Taking $\hat{y}_t$ as the pseudo-label of TD, we can count the numbers of samples in each class and get a pseudo-class distribution. In the data sampler, we assign a selection proportion to every samples in TDs and resample it according to an adaptive proportion calculated by the following formulas:

$$e = -\sum_{i}^{C} \big(\hat{y}_t^i \log \hat{y}_t^i\big) \tag{10}$$

$$p = \frac{1 + \exp(-e)}{N_{\text{class}}} \tag{11}$$

where $\hat{y}_t^i$ is the value of ith bit in the vector $\hat{y}_t$, e is actually the entropy of a prediction vector, and c represents the confidence of a pseudo-label. The higher the information entropy, the higher the prediction uncertainty is. N_{class} is the number of samples in each class. The meaning of the selection probability p is that, the samples from the majority class would have a lower proportion, and for the same class, the samples with higher confidence would have a higher probability to be selected. Therefore, the data sampler can do class rebalancing and screen trusted samples to reduce the impact of low-quality pseudo-labels at the same time

$$\langle \tilde{x}_t, \tilde{y}_t \rangle = \text{DS}\big(x_t, \hat{y}_t\big). \tag{12}$$

The resampled target domain sample set is denoted by $\tilde{x}_t$, the corresponding label is denoted by $\tilde{y}_t$, and DS$(\cdot)$ represents the data sampler.

C. Domain Discriminator

The domain discriminator is responsible for judging whether the sample is from the target domain or a synthetic sample, so as to guide the generator to learn the data distribution of the target domain. The structure of domain discriminator is shown in Fig. 3. It contains four ConvBlocks and two FCBlocks, and the last layer is a single fully connection layer to output a domain prediction. The loss function of domain discriminator is defined as

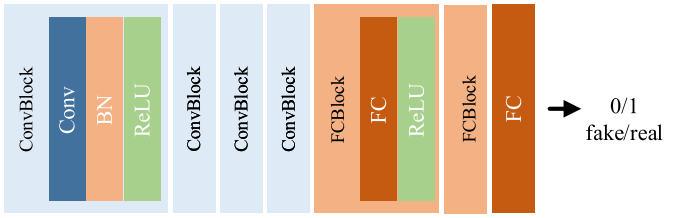


Fig. 3. Structure of domain discriminator.

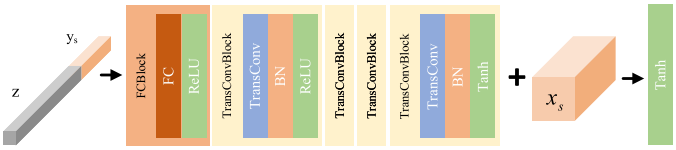


Fig. 4. Structure of generator.

$$L_D = -\Big[\sum \log D(\tilde{x}_t) + \sum \log\big(1 - D(x_f)\big)\Big] \tag{13}$$

where x_f refers to the FD generated by the generator. $D(\cdot)$ is the output of domain discriminator, when the sample is from target domain $D(x) = 1$, else $D(x) = 0$.

D. Generator With Class-Aligned Domain Adversarial Loss

There are two weight sharing generators in CCGDA. The first one is jointly optimized with the classifier to learn the discriminant features in source domain, and the second one aims to learn the target domain distribution with category differences. The generators share the same structure as shown in Fig. 4.

We take y_s and a random noise z as the input of the generator. An FCBlock will map the input vector to a standard dimension and the following four TransConvBlocks will further enlarge the size of generated image. The TransConvBlock is similar to the ConvBlock mentioned before, except that the convolution layer is replaced by the transposed convolution layer. The output of the last TransConvBlock is element-wise added to the SD x_s.

In the joint adversarial optimization with domain discriminator, the generator optimizes itself according to the degree of domain confusion, and in turns, learning the distribution of TD. In the traditional GAN, because the fake samples are generated and put into the discriminator in pairs with the true samples, the categories are aligned, and the data distribution is the only reason that affects the discriminator. However, in DA task, the categories of TD and FD are unaligned, which means that the confusion of domain discriminator may also be caused by the shift of categories. If we also optimize the generator according to this kind of domain confusion, it will lead the generator to learn the distribution characteristics of the target domain with category misalignment.

In CCGDA, an effective way to solve the above problem is to evaluate the category correlation between TD and FD according to the pseudo-label, and then dynamically adjust

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

the optimization direction of the generator according to a new class-aligned domain adversarial loss.

1) Class Correlation Evaluation: The pseudo-labels of TD and FD are obtained by classifier, and the class correlation is calculated based on the CE between the vectors of pseudo-labels. The higher the CE, the higher the class inconsistency between TD and FD, and the lower the class correlation. The calculation process is listed as follows:

$$\mathrm{CE}\left(\tilde{y}_t, \hat{y}_f\right) = -\sum_{i}^{C}\left[\tilde{y}_t^i \log \hat{y}_f^i + \left(1-\tilde{y}_t^i\right)\log\left(1-\hat{y}_f^i\right)\right] \tag{14}$$

$$w = 1 + \exp\left(-\mathrm{CE}\left(\tilde{y}_t, \hat{y}_f\right)\right) \tag{15}$$

where $\tilde{y}_t^i$ and $\hat{y}_f^i$ are the values of ith bit in the pseudo-label $\tilde{y}_t$ and $\hat{y}_f$, respectively, and w is the class similarity finally get.

2) Class-Aligned Domain Adversarial Loss: The loss function of the generator includes two items, a classification loss and a domain discrimination loss. The classification loss is used to train the first generator that is jointly optimized with classifier, while the domain discrimination loss is used to train the second generator that is optimized with the domain discriminator. In order to force the generator to learn the class-differentiated distribution and generate the class-aligned FD, we take the class correlation into account, and define a novel class-aligned domain adversarial loss L_G as follows:

$$L_{\text{class}} = \mathrm{CE}\left(y_s, \hat{y}_f\right) \tag{16}$$

$$L_{\text{domain}} = \sum \log\left(1 - D\left(x_f\right)\right) \tag{17}$$

$$L_G = L_{\text{class}} + \left[1 + \exp\left(-\mathrm{CE}\left(\tilde{y}_t, \hat{y}_f\right)\right)\right] L_{\text{domain}}. \tag{18}$$

E. Training

The training process of CCGDA is shown in Algorithm 1. First, a random noise z, labels of SD y_s and SD x_s are input to the generator to generate FD x_f, then, x_f and x_s are sent to the classifier to get prediction vectors $\hat{y}_f$ and $\hat{y}_s$. Taking y_s as the label of x_f, the classification loss L_C is calculated by (9), and the classifier parameter Θ_C is updated by backpropagation. At the same time, the classification loss L_{class} of the generator is calculated by (16), and the generator parameter Θ_G is updated by backpropagation, forcing the generator to learn the class characteristics of SD. Second, all the TD x_t are sent to the classifier, then the prediction vector $\hat{y}_t$ is obtained as the pseudo-label, and the sample number of each class N_{class} in $\hat{y}_t$ is counted. The sampling probability of each sample is calculated according to (11), and resampling is conducted by the data sampler to achieve the purpose of class rebalancing. Third, the resampled TD $\tilde{x}_t$ and the previously generated FD x_f are sent into the domain discriminator, and the discriminant loss L_D is calculated according to (13) and update the discriminator parameters Θ_D to improve the discriminant ability, while the generator parameters Θ_G are updated according to (17), forcing the generator to learn the distribution of TD by category.

Algorithm 1 Training Process

Input:
the source data x_s, the TD x_t, the labels of source data y_s, a random noise z, the parameters in Classifier, Discriminator, Generator $< \Theta_C, \Theta_D, \Theta_G >$

Output:
the prediction vector of target data $\hat{y}_t$

Begin
for each Epoch **do**
1. joint optimization of generator and classifier
$x_f = Generator(z, y_s, x_s)$
$\hat{y}_f, \hat{y}_s = Classifier(x_f), Classifier(x_s)$
compute the loss L_C by Eq.9
update Θ_C
compute the loss L_{class} by Eq.16
update Θ_G
2. data reasample
$\hat{y}_t = Classifier(x_t)$
Count N_{class} in $\hat{y}_t$
resample the x_t by Data Sampler and output $\tilde{x}_t$
3. joint optimization of generator and discriminator
$d_f, d_t = Discriminator(x_f), Discriminator(\tilde{x}_t)$
compute the loss L_D by Eq.13
update Θ_D
compute the loss L_{domain} by Eq.17
update Θ_G
end for
4. inferring
$\hat{y}_t = Classifier(x_t)$
compute OA, AA and Kappa
End

IV. Experimental Results and Analysis

We conducted experiments on three groups of benchmark datasets to verify the effectiveness of our method, namely the Houston dataset, the HyRANK dataset, and the ShangHang dataset. We select nine DA methods and a pure deep neural network (DNN) without DA technique as comparison algorithms. These nine DA algorithms cover the four branches mentioned in Section II-A. The statistic-based branch includes the following methods: DDC [16], DAN [17], TSTnet [22], joint adaptation network (JAN) [15], and deep subdomain adaptation network (DSAN) [44]. Two representative adversarial-based methods are DANN [33] and MCD [25]. Moreover, a generative-based method PixelDA [18] and a self-training method, debiased self-training (DST) [45] are used. In this section, we will elaborate the experimental process and analyze the results in detail.

A. Dataset Description

1) Houston: The dataset includes Houston 2013 [46] and Houston 2018 [47]. The two images were acquired at the same place of University of Houston campus but at different years (2013 and 2018, respectively) and by different sensors. For the convenience of DA task, 48 bands were selected from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

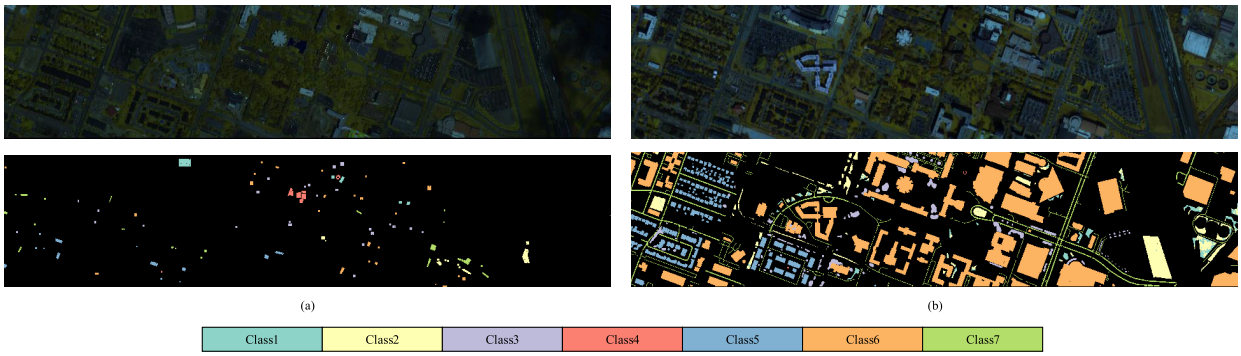


Fig. 5. Pseudocolor image and ground-truth map of Houston dataset. (a) Houston 2013. (b) Houston 2018.

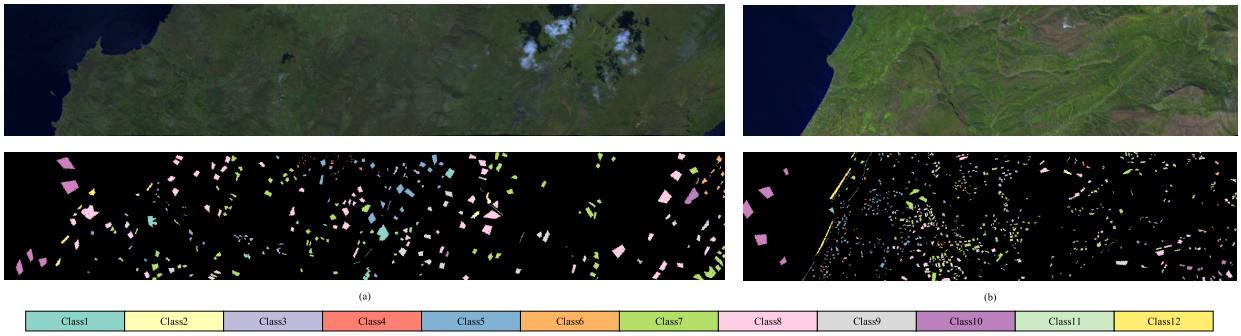


Fig. 6. Pseudocolor image and ground-truth map of HyRANK dataset. (a) Dioni. (b) Loukia.

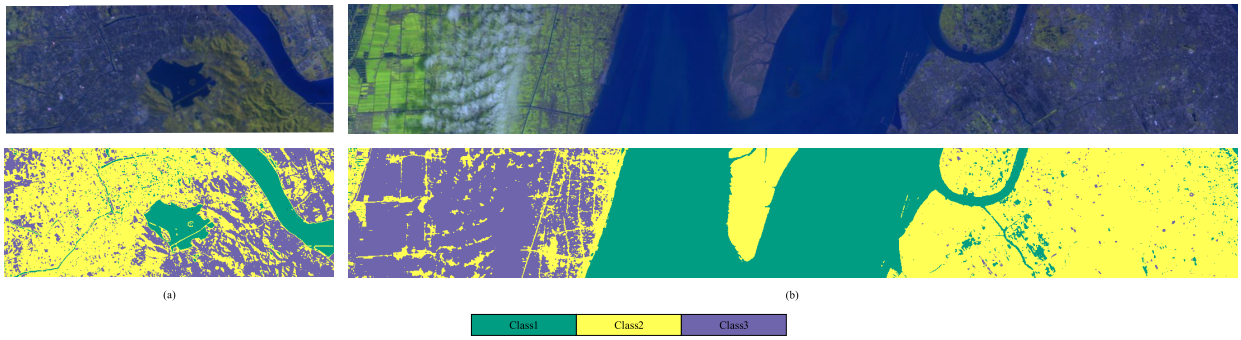


Fig. 7. Pseudocolor image and ground-truth map of ShangHang dataset. (a) Hangzhou. (b) Shanghai.

the original 144 bands in Houston 2013 to match Houston 2018. The size of the processed images of Houston 2013 and Houston 2018 is 210 × 954 pixels with a spatial resolution of 1 m. We take Houston 2013 as the SD and take Houston 2018 as the TD in the experiment. The dataset contains seven categories as shown in Table I. The 46, 26, and 10 bands in HSIs are synthesized into the pseudocolor image as shown in Fig. 5.

2) HyRANK: The dataset includes two scenes of Dioni and Loukia collected by the Hyperion sensor (EO-1, USGS). The size of Dioni is 250 × 1376 pixels, and the size of Loukia is 249 × 945 pixels. Both of them have a spatial resolution of 30 m and a channel dimension of 176. We take Dioni as the SD and take Loukia as the TD in the experiment. The dataset contains 12 categories as shown in Table II, and the differences among the categories are small, which is one of the difficulties of this dataset. The 78, 35, and 10 bands in HSIs are synthesized into the pseudocolor image as shown in Fig. 6.

3) ShangHang: The dataset includes two scenes of Shanghai and Hangzhou collected by the Hyperion sensor (EO-1, USGS). The image size of Shanghai is 590 × 230 pixels, and the size of Hangzhou is 1600 × 260 pixels, both of them have 198 channels. We take Hangzhou as the SD and take Shanghai as the TD in the experiment. The dataset contains three categories as shown in Table III. The 80, 40, and 20 bands in HSIs are synthesized into the pseudocolor image as shown in Fig. 7.

B. Experimental Setting

The parameters and FLOPs (Floating Point Operations) of each network are shown in Table IV. Note that the FLOPs are calculated base on an input size of [1, 48, 27, 27], where 1 is the batch size, 48 is the channel number of Houston dataset, and 27 is the patch size. In the experiments, each sample is represented by its spatial neighborhood patch to extract sufficient spatial information. We set the neighborhood patch

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSES AND NUMBERS OF SAMPLES IN HOUSTON DATASET

Class		Number of Samples	
No.	Name	Houston 2013	Houston 2018
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	total	2530	53200

TABLE II
CLASSES AND NUMBERS OF SAMPLES IN HyRANK DATASET

Class		Number of Samples	
No.	Name	Dioni	Loukia
1	Dense Urban Fabric	1262	288
2	Mineral Extraction Sites	204	67
3	Non Irrigated Arabel Land	614	542
4	Fruit Trees	150	79
5	Olive Groves	1768	1401
6	Coniferous Forest	361	500
7	Dense Sderophyllous Vegetation	5035	3793
8	Sparce Sderophyllous Vegetation	6374	2803
9	Sparcely Vegetated Area	1754	404
10	Rocks and Sand	492	487
11	Water	1612	1393
12	Coastal Water	398	451
	total	20024	12208

TABLE III
CLASSES AND NUMBERS OF SAMPLES IN SHANGHANG DATASET

Class		Number of Samples	
No.	Name	Hangzhou	Shanghai
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
	total	135700	368000

TABLE IV
NETWORK COMPLEXITY

Network	FLOPs(M)	Para(M)
Generator	37.72	0.69
Classifier	80.73	4.25
Discriminator	15.23	1.74

size of each sample to 27×27, and do Z-score normalization before putting it into the network. The specific network parameter settings are shown in Table V, where B refers to the number of spectral bands and is also the length of noise vector z, and C refers to the number of classes.

Ten comparison algorithms are selected, one of which is a common DNN without any DA technique, to show the necessity of DA. The remaining nine algorithms cover the four branches of the DA algorithm mentioned in Section II-A, which makes the comparison algorithm more extensive. Among them, DDC [16] and DAN [17] are classical algorithms in statistic-based branch, and most of the algorithms in this branch are inspired by their essential ideas, so these two methods are widely representative. TSTnet [22] is one of the DA algorithms proposed for HSI in recent years. The method is relatively new and has high performance. Considering that our method is also proposed for HSI, TSTnet is chosen as one of the comparison algorithms. JAN [15] and DSAN [44] solve the problem of category alignment between domains by transforming the MMD loss, while CCGDA also considers the same problem, so they are selected as the comparison algorithm. DANN [33] and MCD [25] are the two most representative algorithms in the adversarial-based branch. The adversarial training of the former one takes place between the feature extractor and the domain discriminator, while that of the latter one takes place between the feature extractor and the two classifiers. PixelDA [18] and DST [45] are the two algorithms that can best represent the generative-based branch and self-training-based branch, so that they are selected as comparison algorithms.

TSTnet is exactly proposed for HSI problem and uses same datasets in their paper for experiments, so we have not adjusted the parameters of it. The other algorithms are originally used in natural images, with a channel number of 3. In order to be used in HSI experiments, we only adjusted their number of channels in convolution layer. For the networks involving feature extractors, we uniformly use AlexNet as the backbone, which ensures that the depth of each network is comparable. For the statistic-based methods except for DAN, their statistical losses are uniformly added to the last FC layer of the classifier. All FC layers in DAN have statistical loss constraints, in which the number of kernel functions of MK-MMD is set to 5. The batch size for all comparison algorithms is 64.

The Adam optimizer is used to optimize the generators, discriminator, and classifier in CCGDA, and the learning rates are set to 0.003, 0.001, and 0.0001 respectively. The batch size is 64 in training process. Our experiment is carried out on the windows system, and the experimental platform is equipped with AMD Ryzen 9 5950X CPU, 128G DDR4 memory, and Nvidia RTX 3090 GPU.

C. Classification Results

The labeled data in SD and the unlabeled data in TD are used to train ten comparison algorithms and our proposed CCGDA method. The well-trained classification models are used to test on TD, and the classification accuracy on each class, overall accuracy (OA), average accuracy (AA), and Kappa coefficients are selected as the metrics of adaptation performance. The higher the values of the above indicators, the better the adaptation performance. We also use the test time as one of the indicators to verify the possibility of practical application of the algorithm.

1) Classification Results on the Houston Dataset: The classification results on Houston dataset are shown in Table VI, and the optimal value of each index is deepened to highlight. It can be observed that, on this hyperspectral dataset,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PARAMETER SETTING OF CCGDA

Net	Layer	In Channel	Out Channel	Kernel Size	Stride	Padding	Output Padding	Output Size
Generator	FC	B+C	(B+C)×16	-	-	-	-	1×1
	reshape	(B+C)×16	(B+C)×4	-	-	-	-	2×2
	TransConv	(B+C)×4	B×4	3	2	1	1	4×4
	TransConv	B×4	B×2	3	2	1	0	7×7
	TransConv	B×2	B	3	2	1	1	14×14
	TransConv	B	B	3	2	1	0	27×27
Classifier	Conv	B	256	3	2	1	-	14×14
	Conv	256	128	3	2	1	-	7×7
	2D-PrimaryCaps	num of capsules = 32						
		128	64	3	2	1	-	4×4
	Conv	128	256	3	2	1	-	4×4
	Conv	256	512	3	2	0	-	1×1
	1D-PrimaryCaps	num of capsules = 32						
		1	1	3	2	1	-	1×1
	DigitCaps	num of capsules = C, digit length = 16, routing times = 3						
Discriminator	Conv	B	256	3	2	1	-	14×14
	Conv	256	128	3	2	1	-	7×7
	Conv	128	256	3	2	1	-	4×4
	Conv	256	512	3	2	0	-	1×1
	FC	512	256	-	-	-	-	1×1
	FC	256	100	-	-	-	-	1×1
	FC	100	C	-	-	-	-	1×1

TABLE VI
CLASSIFICATION RESULTS ON HOUSTON DATASET

class	DNN	DDC	DAN	JAN	DSAN	DANN	MCD	PixelDA	DST	TSTNet	CCGDA
1	41.3 ± 1.4	48.7 ± 3.3	60.7 ± 5.2	42.1 ± 2.1	31.2 ± 3.2	33.8 ± 6.2	42.4 ± 2.5	66.3 ± 4.3	36.8 ± 0.4	53.9 ± 2.1	85.9 ± 1.6
2	72.1 ± 9.1	89.5 ± 4.2	68.3 ± 4.5	33.9 ± 6.1	93.9 ± 0.5	60.4 ± 2.1	61.3 ± 6.6	42.8 ± 3.0	56.0 ± 4.7	78.1 ± 3.1	64.3 ± 5.7
3	45.3 ± 4.7	64.1 ± 11.2	71.2 ± 8.2	61.2 ± 3.2	39.5 ± 3.7	42.7 ± 4.2	57.8 ± 4.5	64.3 ± 1.2	40.9 ± 2.1	58.4 ± 6.8	70.2 ± 1.9
4	1.0 ± 0.1	3.9 ± 0.3	3.2 ± 0.5	3.1 ± 0.1	2.5 ± 0.2	0.1 ± 0.1	1.2 ± 0.4	$68.2\pm$	0.9 ± 0.0	6.1 ± 1.2	81.9 ± 2.1
5	39.5 ± 2.5	38.8 ± 4.2	39.6 ± 0.7	37.8 ± 2.3	53.8 ± 9.8	39.3 ± 7.8	35.0 ± 0.2	59.6 ± 1.0	35.3 ± 7.6	35.2 ± 8.8	88.9 ± 4.1
6	87.4 ± 3.2	85.5 ± 0.8	89.1 ± 1.9	80.1 ± 1.5	76.3 ± 12.0	93.1 ± 2.3	93.1 ± 0.3	75.2 ± 4.4	93.5 ± 1.0	95.3 ± 1.0	82.4 ± 0.1
7	65.3 ± 3.4	62.3 ± 12.3	54.2 ± 2.9	66.2 ± 3.5	53.4 ± 2.9	43.8 ± 2.0	61.5 ± 5.3	45.9 ± 3.8	40.1 ± 11.0	67.9 ± 5.3	63.3 ± 2.6
OA(%)	68.6 ± 0.5	69.4 ± 0.8	68.7 ± 0.1	67.9 ± 0.3	68.3 ± 0.1	59.9 ± 0.7	63.7 ± 0.1	65.9 ± 0.6	59.3 ± 0.2	74.5 ± 0.1	78.1 ± 0.1
AA(%)	55.4 ± 0.4	57.9 ± 1.0	55.1 ± 0.3	48.0 ± 1.4	50.9 ± 0.8	45.0 ± 0.9	51.9 ± 1.7	60.1 ± 0.5	42.3 ± 0.4	57.5 ± 0.4	76.2 ± 0.2
Kappa(%)	55.9 ± 0.7	56.7 ± 0.3	56.4 ± 0.3	53.3 ± 0.2	53.9 ± 0.5	47.9 ± 0.1	52.2 ± 0.3	47.7 ± 0.1	47.1 ± 0.1	61.9 ± 0.4	66.1 ± 0.3
time(s)	6.8	**6.2**	6.7	6.6	6.5	6.9	7.1	6.9	7.1	7.2	6.9

TABLE VII
CLASSIFICATION RESULTS ON HyRANK DATASET

class	DNN	DDC	DAN	JAN	DSAN	DANN	MCD	PixelDA	DST	TSTnet	CCGDA
1	11.4 ± 3.6	13.5 ± 0.4	12.8 ± 2.0	12.7 ± 2.4	9.9 ± 4.3	24.3 ± 3.7	16.0 ± 8.6	31.4 ± 4.4	12.4 ± 5.3	15.0 ± 4.1	45.4 ± 3.2
2	13.1 ± 2.9	13.2 ± 4.5	5.0 ± 1.7	4.0 ± 2.1	0.2 ± 0.2	11.2 ± 6.5	15.9 ± 14.3	49.8 ± 1.8	8.4 ± 3.3	0.1 ± 0.1	39.1 ± 5.3
3	46.8 ± 6.3	45.7 ± 6.3	39.8 ± 1.3	49.6 ± 3.4	61.1 ± 21.7	62.3 ± 7.2	78.0 ± 9.7	46.7 ± 13.2	55.4 ± 25.0	8.4 ± 2.2	77.7 ± 3.3
4	15.2 ± 0.1	19.3 ± 3.9	11.4 ± 4.2	7.9 ± 2.0	11.6 ± 0.2	11.6 ± 3.1	23.3 ± 17.2	44.1 ± 1.2	2.7 ± 2.0	0.0 ± 0.0	48.0 ± 1.1
5	59.6 ± 1.5	60.1 ± 7.1	60.3 ± 5.2	54.1 ± 3.2	50.1 ± 8.3	83.8 ± 2.2	74.6 ± 7.1	24.2 ± 15.2	53.2 ± 9.1	40.6 ± 3.0	64.0 ± 0.2
6	14.3 ± 1.4	25.5 ± 12.1	31.4 ± 13.3	30.8 ± 17.1	29.6 ± 2.5	10.3 ± 3.4	54.2 ± 19.9	5.4 ± 3.8	39.4 ± 24.8	1.2 ± 0.2	13.2 ± 2.7
7	63.4 ± 0.3	69.4 ± 4.7	67.1 ± 2.5	65.0 ± 3.8	63.3 ± 2.5	66.9 ± 2.6	62.9 ± 3.9	51.2 ± 0.6	63.7 ± 4.7	66.4 ± 1.0	63.3 ± 1.5
8	45.6 ± 0.1	44.4 ± 3.8	41.7 ± 2.1	43.2 ± 1.3	38.8 ± 2.3	48.0 ± 1.3	46.8 ± 2.1	14.0 ± 0.9	44.7 ± 2.6	65.3 ± 0.2	59.9 ± 3.3
9	11.9 ± 1.1	15.6 ± 2.3	16.4 ± 1.1	16.0 ± 0.1	15.2 ± 4.5	24.8 ± 2.3	21.2 ± 2.6	67.2 ± 7.7	19.7 ± 2.1	84.1 ± 3.0	79.3 ± 4.5
10	3.1 ± 0.3	27.4 ± 7.0	27.2 ± 3.1	42.5 ± 1.9	33.8 ± 4.1	53.4 ± 15.3	35.3 ± 22.7	61.7 ± 6.2	41.7 ± 18.5	16.2 ± 2.4	56.8 ± 4.1
11	99.9 ± 0.1	99.5 ± 0.2	99.9 ± 0.1	97.6 ± 0.2	98.2 ± 0.5	99.6 ± 0.1	96.9 ± 0.4	99.8 ± 0.0	99.8 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
12	99.8 ± 0.1	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	99.1 ± 0.3	99.9 ± 0.1	96.8 ± 0.9	54.7 ± 2.2	99.8 ± 0.1	100.0 ± 0.0	100.0 ± 0.0
OA(%)	50.4 ± 0.2	52.8 ± 0.3	50.3 ± 0.1	51.6 ± 0.4	49.3 ± 0.1	58.2 ± 0.4	56.1 ± 0.4	41.9 ± 0.3	55.2 ± 0.2	60.6 ± 0.1	63.5 ± 0.2
AA(%)	43.9 ± 0.5	43.5 ± 1.7	42.7 ± 0.8	43.2 ± 2.4	42.9 ± 2.1	49.6 ± 0.4	49.6 ± 0.4	42.2 ± 0.9	47.7 ± 1.9	42.4 ± 0.2	59.4 ± 0.1
Kappa(%)	45.2 ± 0.3	47.3 ± 0.7	45.1 ± 0.2	46.3 ± 0.2	43.9 ± 0.5	53.3 ± 0.3	50.8 ± 0.1	33.5 ± 0.3	50.8 ± 0.4	51.3 ± 0.1	55.7 ± 0.3
time(s)	**4.0**	4.1	**4.0**	**4.0**	**4.0**	4.1	6.2	4.4	4.5	4.7	4.5

conventional DA methods such as DDC, DAN, and JAN do not exhibit significant improvements in adaptation performance. Compared to the DNN without any adaptation enhancement, they only achieve modest improvements of 2.8%, 0.1%, and 1.3% in terms of the OA, respectively. In contrast, both TSTnet and our proposed CCGDA show noticeable accuracy improvements for HSI DA classification tasks. Among the 11 methods, CCGDA achieves the best performance in three metrics: OA, AA, and Kappa. It is worth noting that our method demonstrates a significant advantage in the AA metric, surpassing TSTnet by 18.9%, which is a gain brought by the class-balance function of data sampler. In the Houston

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

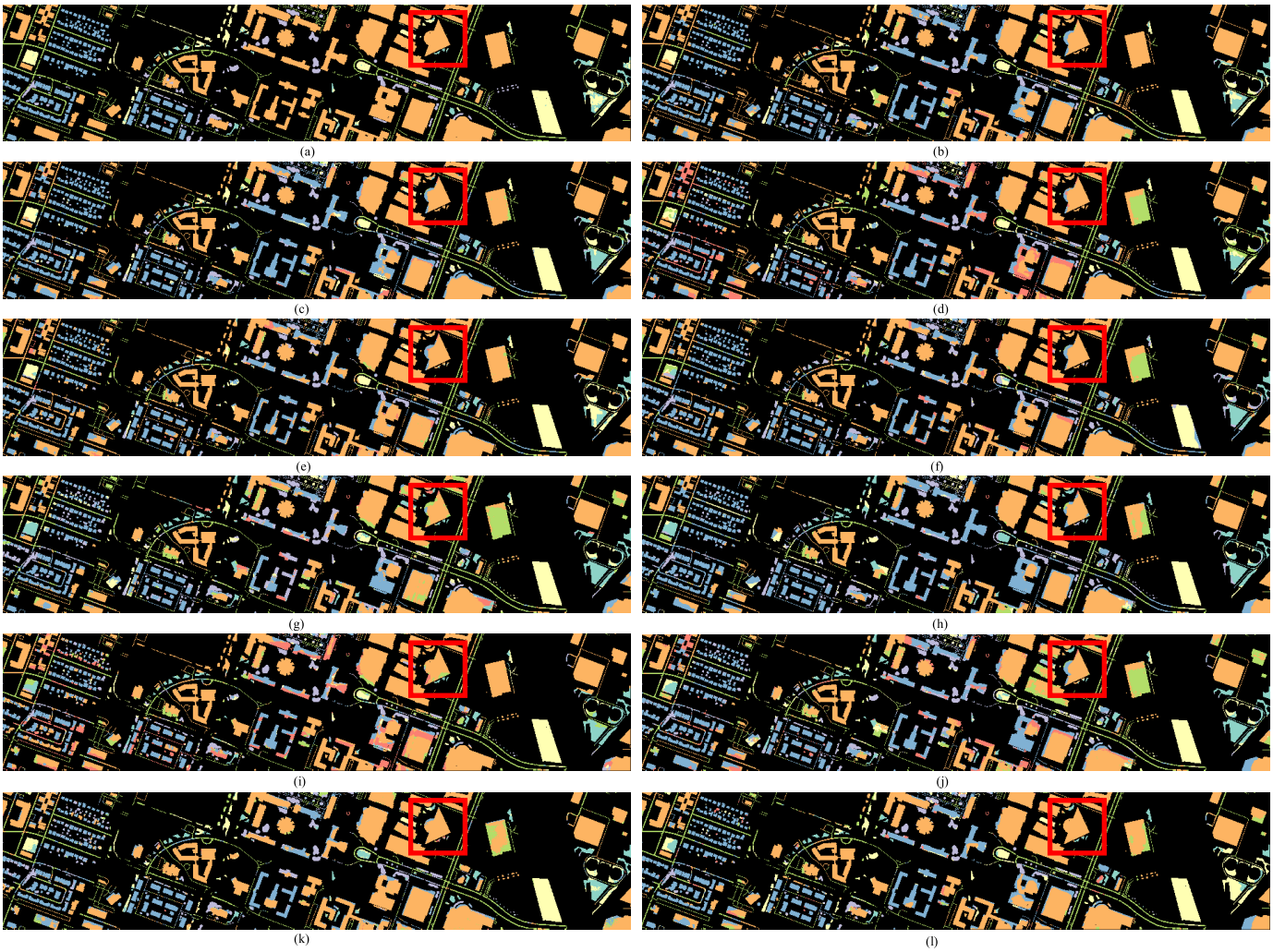


Fig. 8. Classification results of different methods on Houston dataset. (a) Ground truth. (b) DNN. (c) DDC. (d) DAN. (e) JAN. (f) DSAN. (g) DANN. (h) MCD. (i) PixelDA. (j) DST. (k) TSTnet. (l) CCGDA (proposed).

TABLE VIII
CLASSIFICATION RESULTS ON SHANGHANG DATASET

class	DNN	DDC	DAN	JAN	DSAN	DANN	MCD	PixelDA	DST	TSTnet	CCGDA
1	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	99.9 ± 0.1	99.9 ± 0.1	93.4 ± 0.6	100.0 ± 0.0	85.3 ± 1.4	93.9 ± 2.5
2	90.9 ± 0.3	87.4 ± 0.5	91.0 ± 0.81	89.4 ± 1.3	90.5 ± 2.3	88.3 ± 1.0	88.3 ± 2.4	84.9 ± 0.6	88.6 ± 2.3	66.9 ± 3.4	90.3 ± 0.1
3	81.2 ± 0.0	79.5 ± 0.5	79.9 ± 0.4	81.4 ± 2.0	81.7 ± 1.9	83.6 ± 1.1	83.9 ± 1.2	100.0 ± 0.0	84.5 ± 2.3	100.0 ± 0.0	100.0 ± 0.0
OA(%)	90.7 ± 0.0	88.8 ± 0.2	90.6 ± 0.4	89.2 ± 0.1	90.2 ± 0.0	90.5 ± 0.3	85.5 ± 2.1	91.1 ± 0.1	90.5 ± 0.1	78.6 ± 0.3	93.0 ± 0.1
AA(%)	90.7 ± 0.1	89.0 ± 0.2	90.3 ± 0.2	90.0 ± 0.3	90.8 ± 0.1	90.6 ± 0.4	90.6 ± 0.3	92.9 ± 0.2	90.7 ± 0.0	82.6 ± 0.2	94.4 ± 0.1
Kappa(%)	86.6 ± 0.2	83.6 ± 0.3	86.1 ± 0.3	84.7 ± 0.4	86.0 ± 0.5	86.0 ± 0.3	84.7 ± 0.2	86.6 ± 0.2	86.4 ± 0.1	67.7 ± 0.1	89.4 ± 0.1
time(s)	196.5	**191.2**	192.5	197.3	191.6	194.2	192.0	194.6	196.3	197.3	195.4

2018 dataset, the sample size for the fourth class, "water," is only 22. This leads to the sacrifice of accuracy for this class by most algorithms during the feature alignment process, aiming for overall alignment effects. While CCGDA attains 81.8% accuracy on this class, significantly improving the AA. In addition, because the inferring is only carried out by classifier, and the classifier in CCGDA is based on the split-level CapsNet, the network depth is limited. Therefore, inferring on the whole target domain (53 200 samples) only needs 6.9 s.

The classification maps are visualized in Fig. 8. It can be observed that most methods mistake many other categories of regions into class-4 (water), which shows that these methods fail to learn the semantic features of the water category in the target domain because they tend to ignore the minority classes in training process. Compared with other methods, CCGDA shows its good adaptability to the minority samples here.

2) Classification Results on HyRANK Dataset: Our experimental results on HyRANK dataset are shown in Table VII, and the optimal value of each index is deepened to highlight. The classification of vegetation in this dataset is relatively fine-grained, resulting in minimal differences between the various classes. Observing the pseudocolored image in Fig. 6, it is often challenging for the human eye to distinguish between different vegetation classes. This leads to the result that models

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

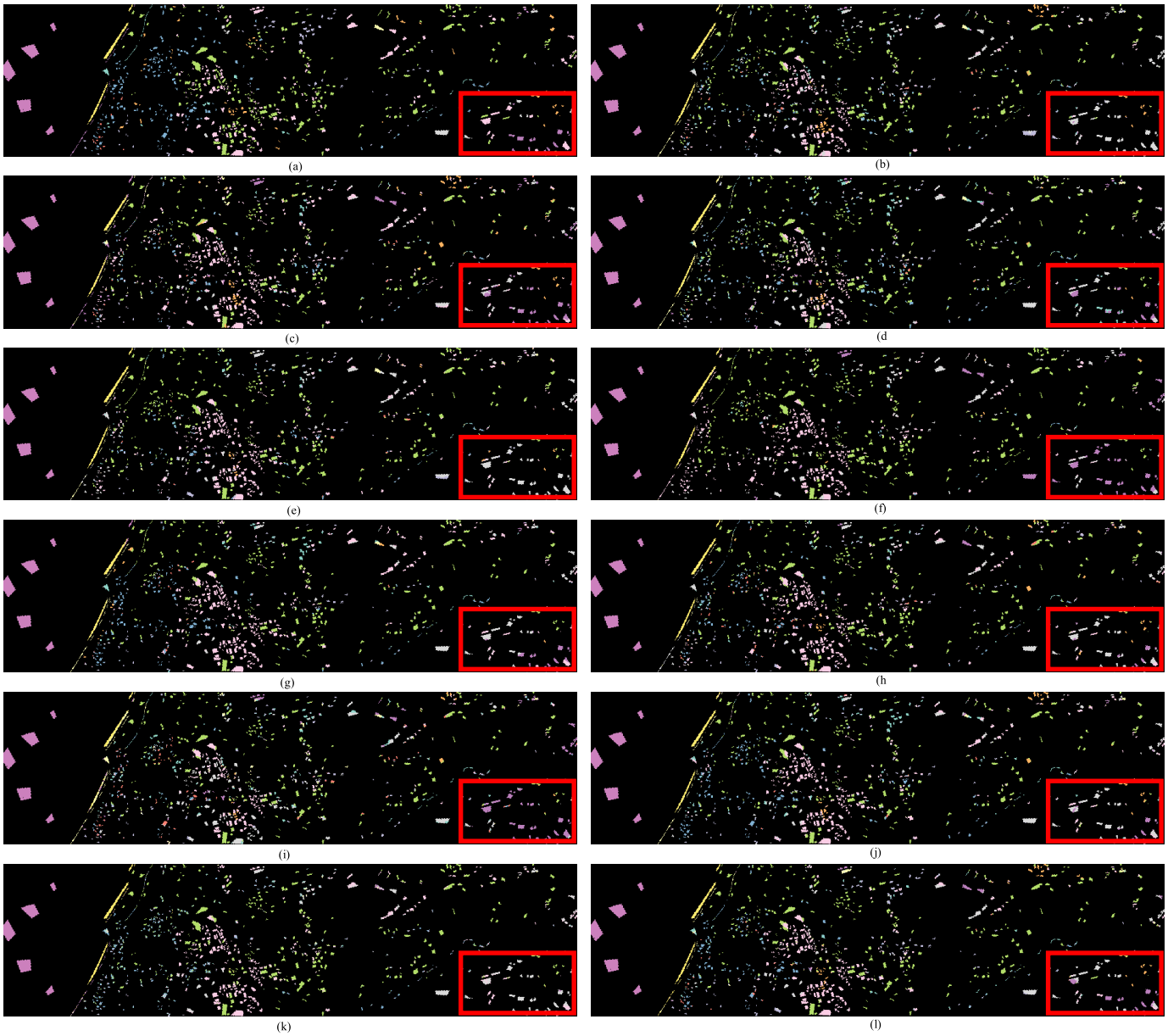


Fig. 9. Classification results of different methods on HyRANK dataset. (a) Ground truth. (b) DNN. (c) DDC. (d) DAN. (e) JAN. (f) DSAN. (g) DANN. (h) MCD. (i) PixelDA. (j) DST. (k) TSTnet. (l) CCGDA (proposed).

often tend to achieve classification performance on the source domain data by overfitting, making them less robust and difficult to transfer to the target domain. Therefore, DA on this dataset poses significant challenges. However, under such conditions, our proposed CCGDA method still achieves an OA of 61.8%, with an AA that is 13.3% higher than TSTnet. The test time of our method on the target domain only takes 4.5 s, which is not significantly different from other algorithms.

The results are visualized in Fig. 9. In the lower right corner of each classification map, most methods mistakenly classify class-10 (rocks and sand) as class-9 (sparsely vegetated area), because the two classes are very similar and the numbers of samples of these two classes in the target domain are small. While CCGDA shows better classification performance in this area.

3) Classification Results on ShangHang Dataset: Our experimental results on ShangHang dataset are shown in Table VIII, and the optimal value of each index is deepened to highlight. Due to the limited number of classes and an ample amount of data, this dataset exhibits good DA performance, with most algorithms achieving an OA of around 90%. Building upon this, our proposed CCGDA method achieves an OA of 93.1% and an AA of 94.2%, which is the best result among these methods. Furthermore, CCGDA achieves an accuracy of over 90% for each class, while other comparison algorithms suffer from notable accuracy deficiencies in certain classes. For example, DDC achieves an accuracy of only 78.8% for the third class, and TSTnet achieves an accuracy of only 62.9% for the second class. Due to the large number of samples in the target domain of ShangHang dataset, the test time of CCGDA reaches

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

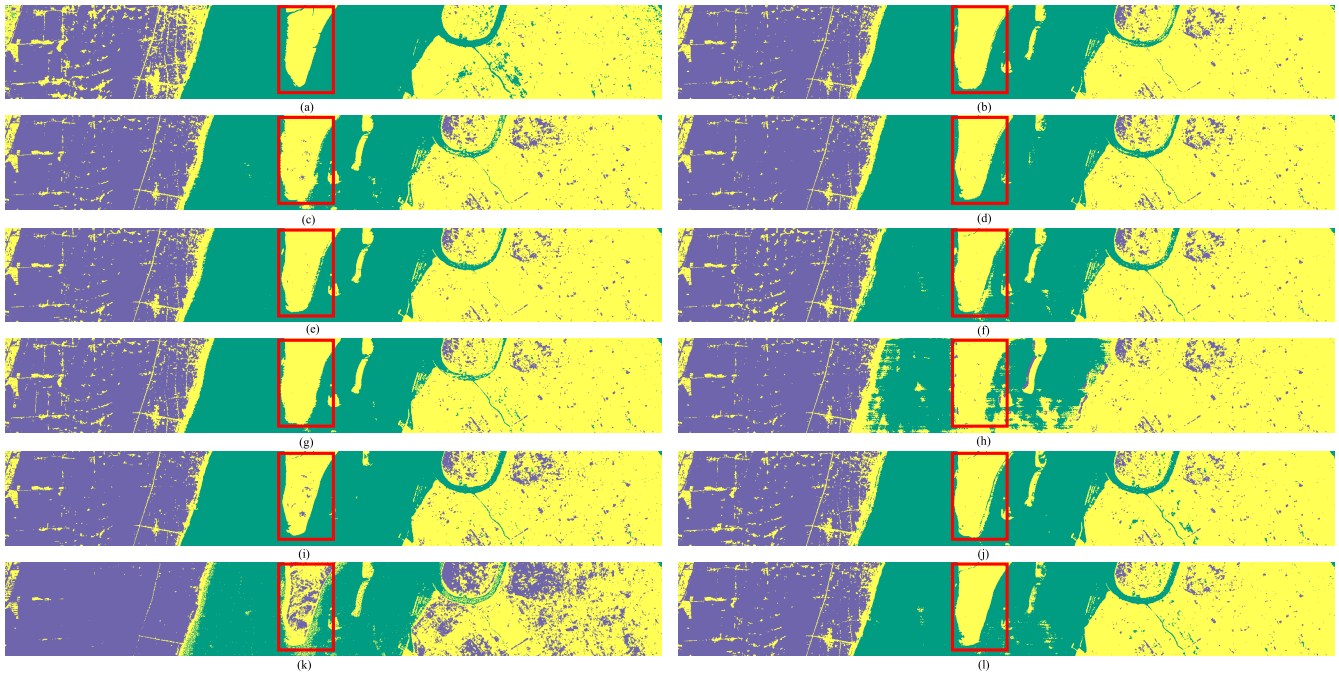


Fig. 10. Classification results of different methods on ShangHang dataset. (a) Ground truth. (b) DNN. (c) DDC. (d) DAN. (e) JAN. (f) DSAN. (g) DANN. (h) MCD. (i) PixelDA. (j) DST. (k) TSTnet. (l) CCGDA (proposed).

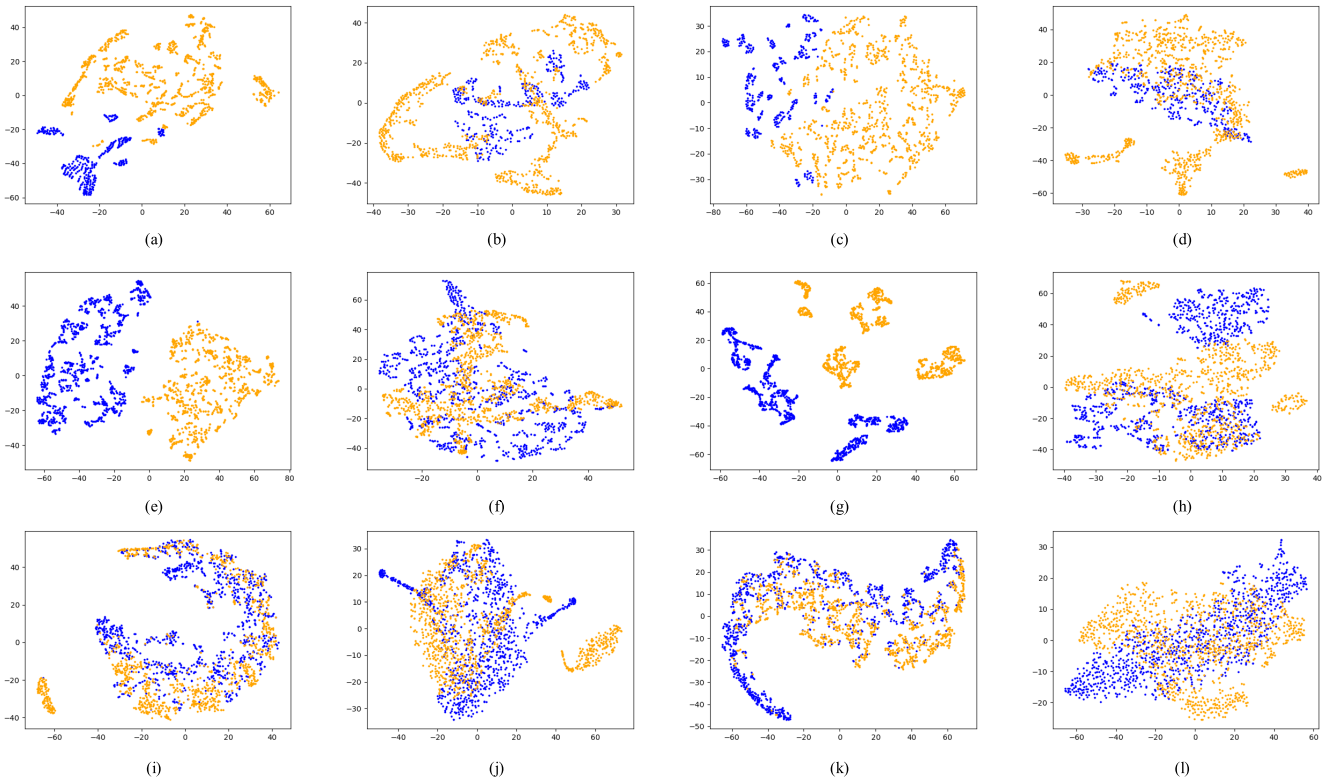


Fig. 11. Feature visualization results of t-SNE dimensionality reduction. Blue dots represent source domain samples, and yellow dots represent target domain samples. (a) Original distribution of class-1 in Houston dataset. (b) Feature distribution of class-1 in Houston dataset aligned by CCGDA. (c) Original distribution of class-3 in Houston dataset. (d) Feature distribution of class-3 in Houston dataset aligned by CCGDA. (e) Original distribution of class-5 in HyRANK dataset. (f) Feature distribution of class-5 in HyRANK dataset aligned by CCGDA. (g) Original distribution of class-11 in HyRANK dataset. (h) Feature distribution of class-11 in HyRANK dataset aligned by CCGDA. (i) Original distribution of class-1 in ShangHang dataset. (j) Feature distribution of class-1 in ShangHang dataset aligned by CCGDA. (k) Original distribution of class-2 in ShangHang dataset. (l) Feature distribution of class-2 in ShangHang dataset aligned by CCGDA.

195.4 s, but there is no obvious difference compared with the other algorithms, which can meet the practical needs.

The results are visualized in Fig. 10. As we can see, CCGDA keeps the edge information and details very well, there is no misclassification noise at the junction of water and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

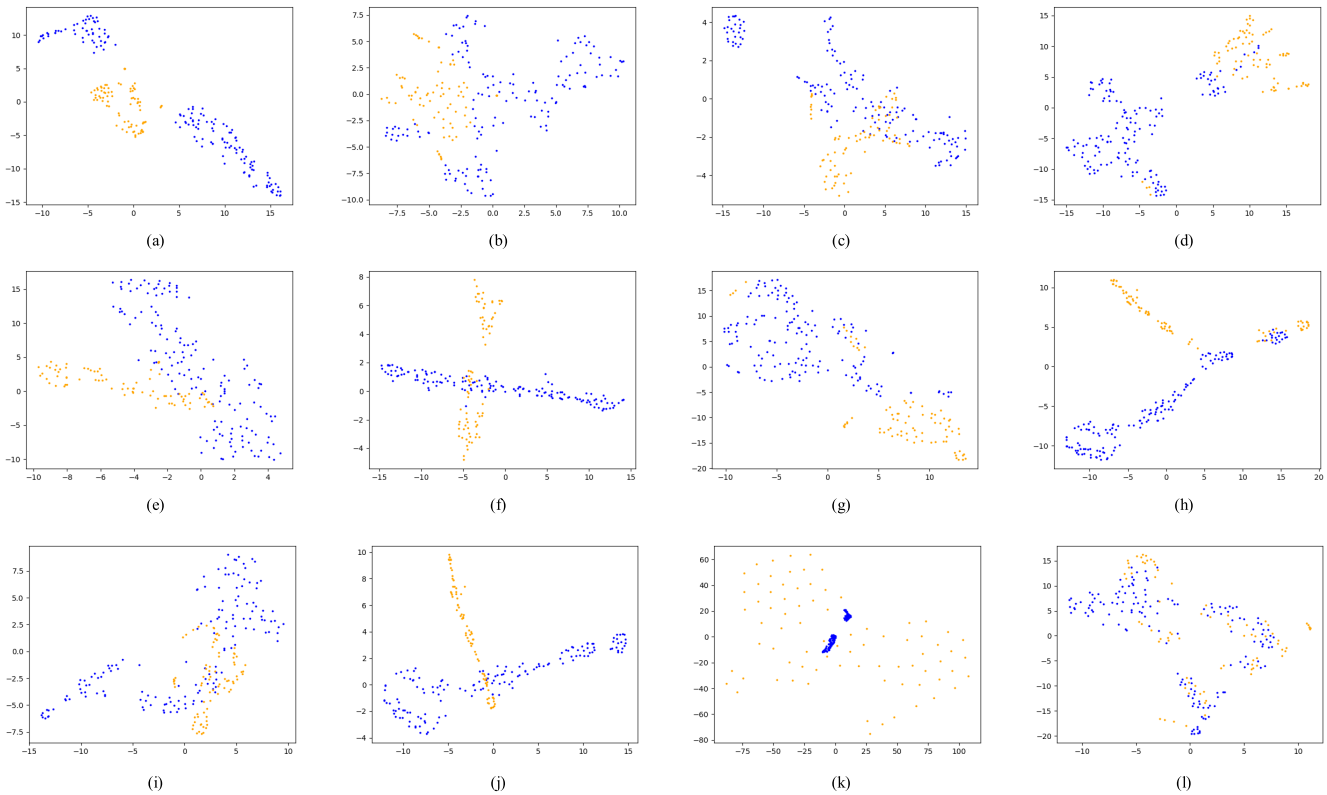


Fig. 12. Feature visualization results of t-SNE dimensionality reduction. Blue dots represent source domain samples, and yellow dots represent target domain samples. (a) Original distribution of class-1 in Houston dataset. (b) Feature aligned by DNN. (c) Feature aligned by DDC. (d) Feature aligned by DAN. (e) Feature aligned by JAN. (f) Feature aligned by DSAN. (g) Feature aligned by DANN. (h) Feature aligned by MCD. (i) Feature aligned by PixelDA. (j) Feature aligned by DST. (k) Feature aligned by TSTnet. (l) Feature aligned by CCGDA.

land compare with the other methods. On the right side of the classification map, it is obviously observed that CCGDA has little misclassification for the class-2 (land/building) and class-3 (plant).

D. Feature Visualization

We use t-SNE method to reduce the dimensionality of the original samples and the aligned features of CCGDA and visualize them, and the results are shown in Fig. 11. Blue dots represent the source domain, and yellow dots represent the target domain. We selected the class-1 and class-3 in the Houston dataset for display, which are both categories with less samples in the target domain. As shown in Fig. 11(a)–(d), where (a) and (c) are the original distribution, (b) and (d) are the feature distribution aligned by CCGDA. From (a) and (c) to (b) and (d), it can be found that the range and center of distribution tend to be consistent, the distribution of features in the two domains overlapped in a large area, indicating that the features of the two domains are aligned. The obvious alignment effect can also be observed on HyRANK dataset in Fig. 11(e)–(h). For ShangHang dataset, we find that the original sample distribution shift is relatively small, as shown in Fig. 11(k), which may be one of the reasons why most algorithms can achieve good adaptation results on this dataset.

Taking class-4 of HyRANK dataset as an example, we compare the aligned features of all algorithms, and the results are shown in Fig. 12. The features of the two domains extracted by different classification network are aligned to a certain extent. It is observed that the sample overlap rate of CCGDA is higher and the distribution boundaries overlap more, indicating that CCGDA has a better performance to some extent.

E. Ablation Experiments

To demonstrate the effectiveness of each component of the CCGDA algorithm, we conducted a series of ablation experiments. The corresponding results are recorded in Table IX. In Table IX, the check mark ($\surd$) indicates that this component is included in the experiment, DS, CCE, and ML refer to the data sampler, class-correlation evaluation, and margin loss, respectively. In addition, we tested the influence of hyperparameters in loss on performance, the results are shown in Table X.

1) CCGDA Without Caps: In this condition, the convolution layer and full connection layer are used to replace the split-level CapsNet in the classifier, so as to ensure that the performance gain brought by split-level CapsNet is verified under the condition of consistent network depth. It can be seen that on the Houston dataset, under the condition of CCGDA w/o caps, OA reduces 11.6%, AA reduces 6.1%, and Kappa reduces 14.2%. It also shows obvious performance degradation on the other two datasets.

2) CCGDA Without DS: This condition deletes the data sampler in the framework, so as to verify the influence of category balance on the adaptation performance. It is observed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
CLASSIFICATION PERFORMANCE IN ABLATION EXPERIMENTS

dataset	Caps	DS	CCE	ML	OA	AA	Kappa
houston	√	√	√	√	**78.3**	**76.2**	**65.6**
	√	√	√		77.2	75.2	64.0
		√	√	√	66.7	70.1	51.4
	√		√	√	71.5	75.2	58.3
	√	√		√	73.3	75.0	59.4
	√			√	69.3	73.2	55.0
		√		√	69.8	66.4	52.3
			√	√	72.1	66.4	53.5
HyRANK	√	√	√	√	**63.8**	**59.4**	**56.3**
	√	√	√		60.0	56.2	52.3
		√	√	√	53.2	49.3	45.3
	√		√	√	58.7	51.3	50.9
	√	√		√	58.6	49.9	50.1
	√			√	57.4	49.9	49.5
		√		√	54.6	48.7	42.9
			√	√	55.5	48.2	43.9
ShangHang	√	√	√	√	**93.1**	**94.2**	**89.4**
	√	√	√		92.1	93.4	88.8
		√	√	√	92.0	93.5	87.8
	√		√	√	90.1	91.8	84.9
	√	√		√	91.8	93.0	87.4
	√			√	91.9	93.3	87.6
		√		√	91.9	93.4	87.7
			√	√	92.7	93.1	88.9

TABLE X
SENSITIVE TO THE HYPERPARAMETER λ

dataset	λ	OA	AA	Kappa
houston	0.1	76.7	74.3	64.1
	0.3	74.4	75.7	62.3
	0.5	**78.3**	**76.2**	**65.6**
	0.7	77.9	75.3	65.5
	0.9	72.7	75.8	59.2
HyRANK	0.1	59.6	57	51.3
	0.3	57.3	48.2	49.1
	0.5	**63.8**	**59.4**	**56.3**
	0.7	57.7	49.3	48.2
	0.9	57.4	49.3	49.6
ShangHang	0.1	92.6	93.8	88.7
	0.3	92.4	93.4	88.8
	0.5	**93.1**	**94.2**	**89.4**
	0.7	92.6	93.8	88.6
	0.9	91.1	92.4	86.3

that under this condition, the performance on the three datasets has obviously decreased.

3) CCGDA Without CCE: This condition cancels the class correlation evaluation between TD and FD in the training process, turning the class-aligned domain adversarial loss into a normal adversarial loss, which does not adjust the optimization direction of the generator according to the category. It also hurts the adaptation performance, just as the significant decline of OA and Kappa indicators on Houston dataset.

4) CCGDA Without ML: This condition keeps the structure of split-level CapsNet, but changes the loss function from margin loss to CE loss in the classifier, which deteriorates the adaptation performance of the algorithm slightly.

5) Sensitive to λ: Most weights of loss in CCGDA are calculated adaptively. Here, we add an experimental discussion on the λ in margin loss. We select five values [0.1, 0.3, 0.5, 0.7, 0.9] for experiments, and we can see that the best result is obtained when λ is 0.5. Too low or too high of the value will lead to a decline in the results.

6) Horizontal Analysis: In CCGDA, the split-level CapsNet has a great influence on the overall adaptation performance of the method, because it is specially designed for the highly generalized feature extraction of HSIs. Replacing split-level CapsNet with convolution network with the same number of layers will lead to the obvious decline of OA, AA, and Kappa indexes. In addition, the class-balance function of data sampler and the class alignment function of class-aligned domain adversarial loss complement each other, deleting either of them will affect the adaptation performance. Compared with the general CE loss, margin loss can improve the confidence of samples, which plays a relatively positive role in the sample filtering process in the proposed framework.

V. CONCLUSION

In this article, a novel DA method (CCGDA) was proposed for HSI classification. In CCGDA, an entropy-constrained class-balancing data sampler helps to avoid the mode collapse caused by majority class preference, and a class-aligned domain adversarial loss guides the generator to learn the target domain distribution with class distinction. A novel split-level CapsNet balances the effectiveness and generalization performance of complex spatial–spectral feature extraction, and explores the possibility of CapsNet in hyperspectral DA task. Experimental results from three benchmark hyperspectral datasets corroborate the superiority of the proposed method, and sufficient ablation experiments verify the effectiveness of each component in this work. This work explores more possibilities of highly robust hyperspectral classification model, and puts forward a new paradigm of generative DA for class alignment in HSIs. In the future, we will pay more attention to the highly generalized discriminant feature extraction for domain adaptive methods under the condition of tiny differences between classes.

REFERENCES

[1] J. Feng, Z. Gao, R. Shang, X. Zhang, and L. Jiao, "Multi-complementary generative adversarial networks with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520018.

[2] J. Wang, S. Guo, R. Huang, L. Li, X. Zhang, and L. Jiao, "Dual-channel capsule generation adversarial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501016.

[3] L. Yan, M. Zhao, X. Wang, Y. Zhang, and J. Chen, "Object detection in hyperspectral images," *IEEE Signal Process. Lett.*, vol. 28, pp. 508–512, 2021.

[4] J. Liang, J. Zhou, L. Tong, X. Bai, and B. Wang, "Material based salient object detection from hyperspectral images," *Pattern Recognit.*, vol. 76, pp. 476–490, Apr. 2018.

[5] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[6] H. Liang, "Advances in multispectral and hyperspectral imaging for archaeology and art conservation," *Appl. Phys. A, Solids Surf.*, vol. 106, no. 2, pp. 309–323, 2012.

[7] G. Lu and B. Fei, "Medical hyperspectral imaging: A review," *J. Biomed. Opt.*, vol. 19, no. 1, Jan. 2014, Art. no. 010901.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.

[8] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.
[9] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based k-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov. 2010.
[10] J. Feng, H. Yu, L. Wang, X. Cao, X. Zhang, and L. Jiao, "Classification of hyperspectral images based on multiclass spatial–spectral generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5329–5343, Aug. 2019.
[11] J. Feng, G. Bai, D. Li, X. Zhang, R. Shang, and L. Jiao, "MR-selection: A meta-reinforcement learning approach for zero-shot hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500320.
[12] J. Feng, N. Zhao, R. Shang, X. Zhang, and L. Jiao, "Self-supervised divide-and-conquer generative adversarial network for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536517.
[13] L. Yu, J. Peng, N. Chen, W. Sun, and Q. Du, "Two-branch deeper graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506514.
[14] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022.
[15] M. Long, H. Zhu, J. Wang, and M. I. Jordan, "Deep transfer learning with joint adaptation networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 2208–2217.
[16] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep domain confusion: Maximizing for domain invariance," 2014, *arXiv:1412.3474*.
[17] M. Long, Y. Cao, and J. Wang, "Learning transferable features with deep adaptation networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 97–105.
[18] K. Bousmalis, N. Silberman, D. Dohan, D. Erhan, and D. Krishnan, "Unsupervised pixel-level domain adaptation with generative adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3722–3731.
[19] M.-Y. Liu and O. Tuzel, "Coupled generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 469–477.
[20] I. Goodfellow, "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
[21] Y. Zou, Z. Yu, B. Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 289–305.
[22] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.
[23] S. Jia et al., "Gradient feature-oriented 3-D domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505517.
[24] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.
[25] K. Saito, K. Watanabe, Y. Ushiku, and T. Harada, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3723–3732.
[26] Y. Ning, J. Peng, Q. Liu, Y. Huang, W. Sun, and Q. Du, "Contrastive learning based on category matching for domain adaptation in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5301814.
[27] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.
[28] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2010.
[29] A. Gretton ET AL., "Optimal kernel choice for large-scale two-sample tests," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1205–1213.
[30] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2016, vol. 30, no. 1, pp. 2058–2065.
[31] B. Sun and K. Saenko, "Deep CORAL: Correlation alignment for deep domain adaptation," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 443–450.
[32] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.
[33] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by back-propagation," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1180–1189.
[34] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 7167–7176.
[35] M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Conditional adversarial domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1640–1650.
[36] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.
[37] Z. Fang et al., "Confident learning-based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527116.
[38] X. Tang, C. Li, and Y. Peng, "Unsupervised joint adversarial domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536415.
[39] Z. Yi, H. Zhang, P. Tan, and M. Gong, "DualGAN: Unsupervised dual learning for image-to-image translation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2849–2857.
[40] J. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2223–2232.
[41] M. Ghifary, W. B. Kleijn, M. Zhang, D. Balduzzi, and W. Li, "Deep reconstruction-classification networks for unsupervised domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 597–613.
[42] C. Wei, K. Sohn, and C. Mellina, "CReST: A class-rebalancing self-training framework for imbalanced semi-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 10857–10866.
[43] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 3856–3866.
[44] Y. Zhu et al., "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, Apr. 2020.
[45] B. Chen, J. Jiang, X. Wang, P. Wan, J. Wang, and M. Long, "Debiased self-training for semi-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 32424–32437.
[46] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[47] B. Le Saux, N. Yokoya, R. Hansch, and S. Prasad, "2018 IEEE GRSS data fusion contest: Multimodal land use classification [technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 52–54, Mar. 2018.



Jie Feng (Senior Member, IEEE) received the B.S. degree from Chang'an University, Xi'an, China, in 2008, and the Ph.D. degree from Xidian University, Xi'an, in 2014.

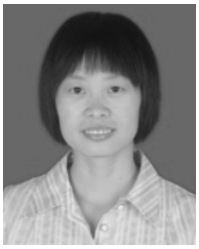
She is currently a Professor with the Laboratory of Intelligent Perception and Image Understanding, Xidian University. Her research interests include remote sensing image processing, deep learning, and machine learning.



Ziyu Zhou received the B.S. degree from Xidian University, Xi'an, China, in 2021, where she is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence.

Her research interests include deep learning and remote sensing image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.



Ronghua Shang (Senior Member, IEEE) received the B.S. degree in information and computation science and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently a Professor with Xidian University. Her research interests include machine learning, pattern recognition evolutionary computation, image processing, and data mining.



Jinjian Wu (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees from Xidian University, Xi'an, China, in 2008 and 2013, respectively.

From 2011 to 2013, he was a Research Assistant with Nanyang Technological University, Singapore, where he was a Post-Doctoral Research Fellow from 2013 to 2014. From 2015 to 2019, he was an Associate Professor with Xidian University, where he has been a Professor since 2019. His research interests include visual perceptual modeling, biomimetic imaging, quality evaluation, and object detection.



Tianshu Zhang received the B.S. degree from Hefei University of Technology, Hefei, China, in 2022, where he is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence.

His research interests include transfer learning and remote sensing image processing.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees in computer application technology from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree in pattern recognition and intelligent system from the School of Electronic Engineering, Xidian University, in 2006.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University. From January 2015 to March 2016, she was a Visiting Scientist with the Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge, MA, USA. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

He has authored or coauthored more than 150 scientific papers. He has charged of about 40 important scientific research projects, and published more than 20 monographs and a 100 articles in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:27:25 UTC from IEEE Xplore. Restrictions apply.